

Client Recommendations Report

Model selection

The held-out KKBox results support selecting XGBoost (Model 2) for further development. This finding is directionally consistent with Matusevičius (2021), whose comparison of Logistic Regression, Decision Tree, Random Forest and XGBoost on KKBox found that XGBoost achieved the highest accuracy among the evaluated models. The present experiment similarly found stronger XGBoost performance, although the numerical results are not directly comparable because the feature engineering, data preparation, experimental splits and evaluation procedures may differ. In this study, XGBoost outperformed Logistic Regression across all reported test metrics (Table 1).

Metric	XGBoost (Model 2)	Logistic Regression (Model 1)
Average precision (AP)	0.8625	0.7104
ROC-AUC	0.9858	0.9669
Precision (at selected threshold)	0.7180	0.6591
Recall (at selected threshold)	0.8453	0.7116
F1-score (at selected threshold)	0.7765	0.6843
Log loss	0.1437	0.3292
Balanced accuracy	0.9113	0.8432
True positives (churners identified)	8 047	6 774
False negatives (churners missed)	1 473	2 746
False positives	3 160	3 504

Table 1

The result is also consistent with Mavridou et al. (2025), who benchmarked several machine-learning approaches on KKBox and reported an AUC of 0.80 for XGBoost compared with 0.68 for Logistic Regression under their 10-fold cross-validation experiment. Their findings provide independent evidence that tree-boosting methods can capture churn patterns in KKBox more effectively than Logistic Regression under some experimental conditions. However, their reported values should not be treated as expected performance for the present experiment. The paired, class-stratified bootstrap estimated test-set metric differences (XGBoost minus Logistic Regression) of +0.1521 for AP (95% CI: 0.1439-0.1596), +0.0189 for ROC-AUC (0.0176-0.0202), and +0.0922 for F1 (0.0858-0.0984). These intervals quantify resampling uncertainty conditional on the test set and fitted models; they do not account for model-training or future-population uncertainty.

In this project, XGBoost achieved higher AP, ROC-AUC, recall, F1-score and balanced accuracy, together with lower log loss. The paired bootstrap confidence intervals for AP, ROC-AUC and F1 were entirely positive, supporting the observed differences on the held-out test set. These results, rather than the literature alone, form the basis for selecting XGBoost. The recommendation is therefore to carry XGBoost forward as the leading candidate, not to deploy it immediately at STADIOchoice. The models were trained and tested on the public KKBox music-streaming churn

dataset, and this consequently establishes comparative performance on KKBox only and does not establish expected performance and operational suitability for STADIOchoice.

Improving the selected model

Hyperparameter tuning and model stability

A more systematic hyperparameter search should be conducted, using training data for fitting and validation data for model and threshold selection. The current experiment explored only a limited range of tree depths and learning rates. Thus, further work should also consider regularisation, minimum child weight, number of estimators, and row and column subsampling, while monitoring computational cost. This approach is consistent with Mavridou et al. (2025), who benchmarked multiple configurations using cross-validation and weighed predictive performance against training and inference efficiency, showing that model selection must balance accuracy with the resources required to achieve it. To assess stability, cross-validation or repeated stratified splits should be used within the development data, while the held-out test set is preserved for final evaluation so that it does not influence further model or threshold choices.

Business-aligned evaluation and threshold selection

Evaluation should retain AP, precision, recall, F1-score, balanced accuracy and log loss as complementary metrics rather than relying on a single measure. Mavridou et al. (2025) evaluated churn models using AUC, recall and log loss in addition to accuracy, demonstrating the value of assessing different aspects of predictive performance. Within this framework, the classification threshold should be set with commercial and customer-operations stakeholders, explicitly trading off the cost of missed churners (lost subscriptions and lifetime value) against unnecessary outreach and intervention capacity, instead of simply maximising F1-score. Although Mavridou et al. (2025) emphasise efficient targeting because retention campaigns incur costs, their subscription-software setting does not determine the organisation's actual intervention costs or capacity and thus, these business-specific values must be obtained internally before a defensible threshold can be set. Finally, probability calibration should be assessed before interpreting predicted probabilities as churn risks, specifically by evaluating the log loss (already reported in Part C) which summarises overall probability quality, but at times a lower value does not guarantee well-calibrated risks for the target subscriber base.

Model validity and governance

Model validity requires verifying that all predictors would be available at the intended prediction time and that no information from after that date leaks into training, including confirming the temporal alignment between KKBox feature windows and the churn-label period. Cheng and He (2025) aggregated transaction histories and listening behaviour to construct user-level features before combining them with churn labels. Their work illustrates the value of multi-source feature construction but does not imply that every aggregated feature is suitable for prospective prediction without checking its availability date. Beyond temporal correctness, subgroup performance, missing-data patterns, model stability and potential data drift should be assessed, and the model's limitations and appropriate use conditions documented. Feature importance can help identify influential predictors, but should not be interpreted as evidence that changing a feature will prevent churn: Cheng and He (2025) highlight behavioural and payment-related variables as important, yet their predictive analysis does not establish causal effects of retention interventions.

Adapting the model to STADIOchoice data

Data assessment and target definition

The KKBox feature schema cannot be transferred directly to STADIOchoice. Once the SS1-requested data are received, the first step is to review the data dictionary and assess coverage, missingness, categorical encodings, effective dates and observation grain. In parallel, the definition of customer churn must be agreed with business stakeholders, specifying the cancellation or termination event, prediction date and any exclusions. Because STADIOchoice offers satellite subscriptions, STADIOstream subscriptions and households using both services, the churn definition should distinguish between these segments and reflect the organisation's business rules.

Customer-level modelling dataset

A modelling table should be constructed with one record per eligible customer or household per prediction date, depending on data availability and the agreed prediction objective. Predictors must be derived using only information available on or before the prediction date, while churn labels are defined from outcomes within the subsequent agreed period. Relevant predictors may include subscription history, payment behaviour, viewing activity and service usage, subject to what the data can support. Cheng and He (2025) demonstrate the use of behavioural and transaction-derived features for KKBox churn prediction, but STADIOchoice's feature definitions, observation grain and prediction horizon must be established from its own data and business requirements. Where multiple observations per customer are used, chronological splits should be applied to prevent records from the same customer or household from leaking across training and evaluation sets, and all preprocessing transformations should be fitted on training data only.

Model redevelopment and validation

With the STADIOchoice dataset in place, XGBoost should be retrained and tuned on the training data, with class weights recalculated and the classification threshold selected using validation data. XGBoost should then be compared against Logistic Regression using identical data splits, target definitions and evaluation metrics, following the comparative approach of Matusevičius (2021) and Mavridou et al. (2025). Where sufficient historical data exist, both models should additionally be evaluated on a later period to assess temporal stability before any operational use. The KKBox results should not be presented as forecasts of STADIOchoice churn performance but should be seen as a methodological benchmark on a different domain.

Alignment with Part A literature

The findings align broadly with existing literature while remaining specific to this experiment. Matusevičius (2021) and Mavridou et al. (2025) both demonstrated XGBoost's superior performance over Logistic Regression on KKBox churn data, which is consistent with the results, although methodological differences preclude direct numerical comparison. Meanwhile, Cheng and He (2025) highlighted graph-attention architectures as an alternative feature-fusion approach, though adapting such models would require comparable relationship data.

Overall, Part C supports advancing XGBoost to the next proof-of-concept stage. The primary objectives are to validate the model on properly defined STADIOchoice student data against a robust baseline under identical conditions and to determine whether its predictions can effectively underpin feasible retention interventions.

References

Cheng, H. and He, J. (2025) 'Advanced customer churn prediction for a music streaming digital marketing service using attention graph-based deep learning approach', Scientific Reports, 15, 45028. <https://doi.org/10.1038/s41598-025-28357-z>

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